NEXA
TECHNOLOGY FOR A BRIGHTER TOMORROW

USER GUIDE
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NexaAI Mini Workstation User Guide

Compact. Powerful. Built for Local AI.



1. What's in the Box


NexaAI Mini Workstation


Power Adapter (120W)


HDMI Cable (1.5 m)



Quick Start Guide

2. Technical Specifications

Specification	Details
Model Number	NS-MINI-AI-2026
Processor	Intel® Core™ Ultra 7 155H (16 cores, up to 4.8 GHz)
NPU	Intel® AI Boost (up to 11 TOPS)
Graphics	NVIDIA® GeForce RTX™ 4060 Laptop GPU (8 GB GDDR6)
Memory	32 GB DDR5 (5200 MHz), expandable to 64 GB
Storage	1 TB NVMe PCIe 4.0 SSD, expandable (M.2 2280)
Connectivity	2 × USB-A 3.2 Gen 2, 2 × USB-C (Thunderbolt™ 4), HDMI 2.1, DisplayPort 1.4, 2.5G LAN, Wi-Fi 7, Bluetooth 5.4
Video Output	Up to 4 displays (1 × HDMI 2.1, 1 × DP 1.4, 2 × USB-C)
Operating System	Windows 11 Pro / Ubuntu 24.04 LTS (optional)
Dimensions	155 × 155 × 52 mm
Weight	1.2 kg
Power Supply	120W external adapter

3. Ideal Use Cases

Local LLM Inference	AI Development and RAG	Docker & Kubernetes	Software Development	Home Lab and Learning

1. Product Overview

NexaAI Mini Workstation is a compact desktop for local AI application development, RAG prototyping, software engineering, home labs and containerized workloads. It is positioned between a developer laptop and a full tower AI workstation.

2. Technical Specifications

Specification	Details
Processor	Intel Core Ultra 7-class mobile processor, 16 cores
NPU	Integrated AI engine, up to 11 TOPS
Graphics	NVIDIA GeForce RTX 4060 Laptop-class GPU, 8 GB GDDR6
Memory	32 GB DDR5-5200; expandable to 64 GB
Storage	1 TB PCIe 4.0 NVMe SSD; second M.2 slot
Networking	2.5 GbE, Wi-Fi 7, Bluetooth 5.4
Display support	Up to four displays through HDMI / DisplayPort / USB-C
Ports	USB-C / Thunderbolt-class port, USB-A, HDMI, DisplayPort, audio
Power	120 W external adapter
Dimensions	Approx. 155 x 155 x 52 mm
Weight	Approx. 1.2 kg
OS	Windows 11 Pro; selected Linux distributions may be used where drivers are supported

3. Ideal Use Cases

Use case	Suitability	Notes
Cloud AI / API development	Excellent	Compact always-on development machine for API-based AI solutions.
RAG / embeddings	Excellent	Good for local vector databases, ingestion services and embedding experiments.
Small local LLMs	Good	8 GB VRAM supports selected quantized models; model/runtime requirements vary.
Docker development	Excellent	Suitable for multiple local services when memory is sized appropriately.
Local Kubernetes labs	Good	Useful for compact learning/dev clusters; monitor memory pressure.
Computer vision	Good	Appropriate for prototyping and moderate GPU workloads.
Large-model training	Not primary use	Use a full workstation or cloud GPU resources for heavier workloads.

4. Setup

Connect display, keyboard, mouse and network before first boot. Keep at least 75 mm of clearance around ventilation areas. Install operating-system updates, current GPU drivers, development SDKs and approved local AI runtimes before beginning sustained AI workloads.

5. Developer Environment

Layer	Typical tools
Application development	.NET SDK, Visual Studio / VS Code, Git
Containers	Docker or compatible container runtime
Local orchestration	kubectl and Helm where required
Data	PostgreSQL, Redis and local development databases
AI	Model-provider SDKs, Microsoft.Extensions.AI, local inference runtime
RAG	Vector database, embedding pipeline and ingestion worker

6. Troubleshooting

Symptom	Recommended checks
Local model will not load	Check available GPU VRAM, system RAM, model quantization and runtime compatibility.
GPU not visible in container	Verify host driver, GPU container integration and runtime configuration.
System throttles under load	Check ventilation, ambient temperature and performance profile.
Storage fills quickly	Move model files to secondary NVMe storage or remove unused model versions.
Multiple containers become slow	Check memory pressure, CPU saturation and background services.

Educational notice

NexaAI Mini Workstation is a fictional product created for the NexaStore educational case study.